project.org

STUDENT HANDOUT

RED LEVEL
GO DEEPER

Brain Detective

Grades 1–2

ADMIT ONE

NAME _____ DATE _____ CLASS _____

Your brain receives 11 million pieces of information every second through your eyes, ears, and other senses. You can only pay attention to about 40 of them! So your brain makes predictions to fill in the gaps. Usually this helps you—but sometimes your predictions are wrong.

Amazing Fact: Your brain is so good at predicting that you can read scrambled words, see things that aren't really there, and even fill in parts of pictures your eyes can't actually see!

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PART 1

Create Your Own Brain Trick

Now that you know how brains predict, can you create something that tricks someone's brain?

Ideas:

- Write a scrambled sentence that's really hard to read
- Create a "spot the difference" picture
- Draw something that could be two different things

Test it on a friend or family member. Did their brain make the wrong prediction?

What happened? _____

CREATE YOUR BRAIN TRICK HERE!

02

PART 2

Real-Life Brain Detective Story

Write about a time when you (or someone you know) made a wrong prediction and had to check to find out the truth:

What was the wrong prediction? _____

What was the truth? _____

How did you find out? _____

WRITE OR DRAW YOUR STORY HERE

03

PART 3

Think Deeper: Challenge Questions

1. Why do you think our brains make predictions instead of seeing everything exactly as it is?

2. When are predictions helpful? When are they NOT helpful?

3. How can you train yourself to be a better Brain Detective?

MINI MUSEUM CONNECTION

Mini Museum Challenge. When you visit the Mini Museum (or explore online at minimuseumproject.org/exhibitions/seeing-is-deceiving/illusions), pay special attention to the Victorian illusion cards. These are over 100 years old!

What prediction do they make your brain create? How do they work?

SHARE WITH YOUR FAMILY

Family Research Project: With your family, find 3–5 optical illusions online. For each one, try to figure out: What prediction is your brain making? Why does the illusion work?

Create a “Brain Detective Guide” poster showing your favorite illusions and explaining how they trick your brain!